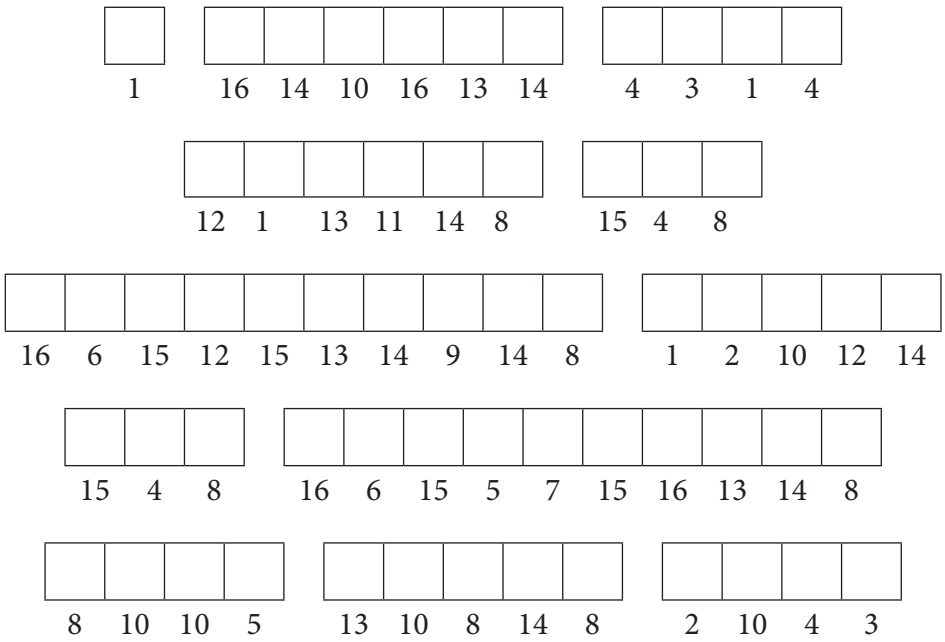
Name: _____

WORKSHEET

Sketching periodic functions – phase and vertical shift



Dwight D. Eisenhower

Rule

Given a periodic function $y = \cos(x + C) + D$

C is the **phase shift**

D is the **vertical shift**

$+C$ moves the function C *to the left.*

$-C$ moves the function C to the right.

$+D$ moves the function up D.

$-D$ moves the function down D.

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Questions

1 What is the phase shift of $y = \sin(x + \pi)$?

2 Sketch this function.

3 What is the vertical shift of $y = \tan(x - 1)$?

4 Sketch this function.

5 What is the phase shift of

$$y = \cos\left(x - \frac{\pi}{2}\right)?$$

6 Sketch this function.

7 What is the vertical shift of $y = \sin(x + 2)$?

8 Sketch this function.

For the function

$$y = \tan\left(x + \frac{\pi}{2}\right) - 2:$$

9 State the phase shift.

10 State the vertical shift.

11 Sketch this function.

For the function

$$y = \cos(x - \pi) + \frac{1}{2}:$$

12 State the phase shift.

13 State the vertical shift.

14 Sketch this function.

15 Sketch the graph of $y = \sin(x + 2\pi) - 1$.

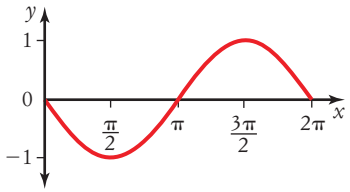
16 Sketch the graph of

$$y = \cos\left(x - \frac{\pi}{4}\right) + 3.$$

Solutions

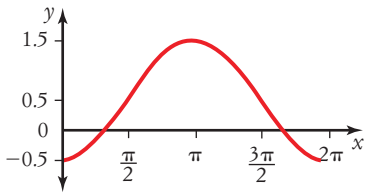
A π to the left

B



C 2 up

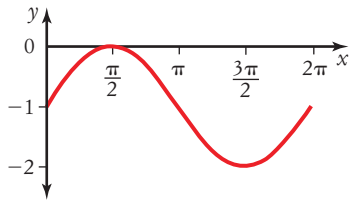
E



G $\frac{\pi}{2}$ to the left

H 1 down

I

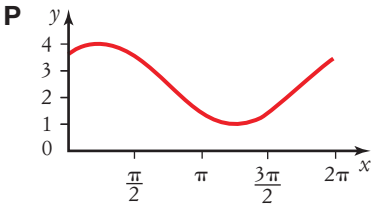


L $\frac{1}{2}$ up

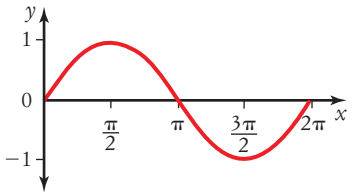
N $\frac{\pi}{2}$ to the right

O 2 down

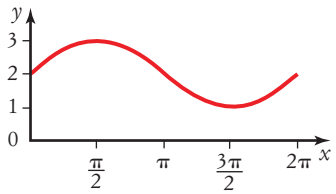
P



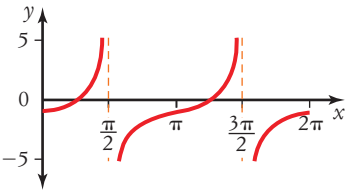
R



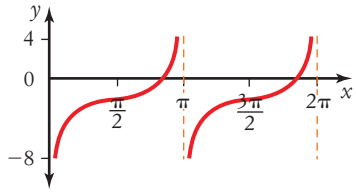
S



T



U



V π to the right

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